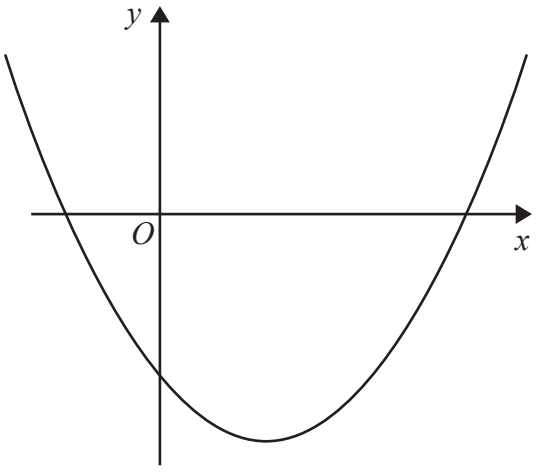
17 Here is a sketch of a curve.



The equation of the curve is $y = x^2 + ax + b$ where a and b are integers.

The points $(0, -5)$ and $(5, 0)$ lie on the curve.

Find the coordinates of the turning point of the curve.

At $(0, -5)$: $-5 = 0 + 0 + b$, so $b = -5$.

1 mark
Substitute a given point

At $(5, 0)$: $0 = 25 + 5a - 5$

$5a = -20$, so $a = -4$.

1 mark
Use the other point to find a

$y = x^2 - 4x - 5$; factors of -5 with sum -4: -5 and 1.

	x	-5
x	x^2	$-5x$
1	x	-5

$y = (x - 5)(x + 1)$: roots are 5 and -1.

Turning-point $x = \dfrac{5 + (-1)}{2} = 2$

1 mark
Midpoint of the roots

At $x = 2$: $y = 4 - 8 - 5 = -9$.

1 mark
Both coordinates found

(2 , −9)

(Total for Question 17 is 4 marks)